Consider the double-slit experiment, which illustrates so vividly how quantum phenomena are simultaneously wave-like (we see interference patterns) and particle-like (we see dots on the detector screen, and interference goes away when we detect which slit the particles go through). In Bohmian mechanics this ambiguity is not mysterious at all: there are both particles and waves. The particles are what we observe; the wave function affects their motion, but we have no way of measuring it directly.

According to Bohm, the wave function evolves through both slits just as it would in Everettian quantum mechanics. In particular, there will be interference effects where the wave function adds or cancels once it reaches the screen. But we don't see the wave function at the screen; we see individual particles hitting it. The particles are pushed around by the wave function, so that they are more likely to hit the screen where the wave function is large, and less likely to do so where it is small.

The Born rule tells us that the probability of observing a particle at a given location is given by the wave function squared. On the surface, this seems hard to reconcile with the idea that particle positions are completely independent variables that we can specify as we like. And Bohmian mechanics is perfectly deterministic—there aren't any truly random events, as there are with the spontaneous collapses of GRW theory. So where does the Born rule come from?

The answer is that, while in principle particle positions could be anywhere at all, in practice there is a natural distribution for them to have. Imagine that we have a wave function and some fixed number of particles. To recover the Born rule, all we have to do is start with a Born rule–like distribution of those particles. That is, we have to distribute the positions of our particles so that the distribution looks like it was chosen randomly with probability given by the wave function squared. More particles where the amplitude is large, fewer particles where it is small.

Such an "equilibrium" distribution has the nice feature that the Born rule remains valid as time passes and the system evolves. If we start our particles in a probability distribution that matches what we expect from ordinary quantum mechanics, it will continue to match that expectation going forward. It is believed by many Bohmians that a non-equilibrium initial distribution will evolve toward equilibrium, just as a gas of classical particles in a box evolves toward an equilibrium thermal state; but the status of this idea is not yet settled. The resulting probabilities are, of course,